Speech-Native Retrieval-Augmented Generation for Agricultural Advisory in Low-Resource Dialects

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Abstract*—Smallholder farmers in low-literacy, dialect-rich regions face a persistent barrier to digital agricultural advisory: existing voice-enabled systems rely on a cascaded pipeline in which automatic speech recognition (ASR) first transcribes a spoken query before retrieval-augmented generation (RAG) produces an answer. Because domain-critical terminology — pesticide names, crop diseases, and pest names — is disproportionately subject to ASR substitution errors, and because farmers frequently use dialect-specific or colloquial terms absent from standard ASR training data, transcription errors propagate directly into retrieval failures and, ultimately, into incorrect agricultural advice. Recent work has shown that retrieval can be performed directly on speech embeddings without an intermediate transcription step, removing one source of cascade error; however, no existing system addresses the compounding effect of dialect variation and domain-critical rare vocabulary together, nor combines speech-native retrieval with an explicit dialect-to-scientific-entity mapping layer, confidence-gated escalation to a human expert, and fully offline operation on resource-constrained hardware. This paper proposes a framework that integrates these four components and outlines an experimental protocol — comparing ASR-cascaded, keyword-based, dense semantic, and hybrid retrieval under dialect variation — to evaluate whether this combination improves retrieval accuracy and answer reliability for agricultural advisory in low-resource dialect settings. A runnable software prototype of the proposed pipeline was built and measured end to end on small, demo-scale data as a preliminary mechanism check. On ground-truth query text, dialect mapping is what makes dialectal queries retrievable at all in this setting, and per-stage latency fits comfortably within a consumer 6 GB-VRAM budget in the pipeline's fp16 configuration; on synthesized speech audio, however, neither the ASR-cascaded nor the speech-native retrieval condition performs well, leaving this paper's central speech-native premise unconfirmed on this data, with the speech-native adapter's failure to generalize past its training distribution remaining unresolved pending evaluation on real target-dialect speech, which does not yet exist. Two further follow-up measurements were added afterward: re-testing the confidence gate against real, non-template-quoting LLM-generated answers traced its first validation's apparent open-book-signal weakness to an artifact of that validation set rather than a fundamental flaw, improving the gate's measured performance; and quantizing the local language model to INT8/INT4 unexpectedly *increased* per-query latency several-fold despite reducing memory use, qualifying the offline-deployment finding to the fp16 configuration actually benchmarked.

***Keywords*—Agricultural advisory systems, dialect-aware retrieval, low-resource languages, speech-native RAG.**

I. Introduction

Agricultural extension — the transfer of evidence-based farming knowledge to smallholder farmers — remains a critical bottleneck in regions with low literacy and limited connectivity. RAG-based advisory systems such as Farmer.Chat [1] and KrishokBondhu [2] show that voice-enabled, multilingual chatbots can reduce farmers' dependence on scarce extension agents, and systems such as SukhaRakshak AI [3] extend this to more than twenty Indian languages. All of them, however, first transcribe a spoken query via automatic speech recognition (ASR) before retrieval proceeds on the resulting text — a cascaded design vulnerable to a specific failure mode. ASR errors on Indian-language agricultural speech disproportionately affect domain-critical terms such as crop, chemical, and pest names [4], and this degrades further under dialectal and code-mixed variation [5]. A mistranscribed pest or disease term propagates directly into retrieval, surfacing an irrelevant document and ultimately an incorrect recommendation.

One way to remove this failure mode is to retrieve directly from a speech representation, bypassing transcription entirely — a direction that has recently moved from proposal to validated, production-deployed technique. Amazon's SpeechRAG [6] aligns a speech encoder with a frozen text-retrieval embedding space to retrieve audio passages without ASR, and Google's Speech-to-Retrieval (S2R) [7] performs an analogous mapping live in production for Voice Search. Neither system, however, has been evaluated under dialect variation or domain-critical rare-entity retrieval such as pesticide and pest names, and neither normalizes the many regional or colloquial terms a farmer may use for the same concept. Existing agricultural knowledge graphs [8], [9] normalize only standardized, written terminology, leaving vernacular-to-scientific mapping for spoken queries an open problem.

A further limitation is that existing systems answer regardless of confidence, even though incorrect agricultural advice can damage a season's crop — an argument made explicitly for comparably high-stakes RAG deployments in clinical and financial domains [10], [11], though naive confidence signals are known to misfire in both directions [12] and require validation rather than blind trust. Most RAG research also implicitly assumes reliable connectivity and cloud-hosted inference [13], which the rural settings these systems target frequently lack. Taken together, no existing system combines speech-native retrieval, dialect-to-entity mapping, confidence-gated escalation, and offline operation for agricultural advisory in low-resource dialects. This paper addresses that combination and poses the following research questions:

- **RQ1:** Does speech-native retrieval retain its advantage over ASR-cascaded retrieval under dialect variation and domain-critical rare vocabulary?
- **RQ2:** Does a dialect-to-scientific-entity mapping layer improve retrieval accuracy for queries phrased in regional or colloquial terms?
- **RQ3:** Can the pipeline operate fully offline, on resource-constrained consumer hardware, at acceptable latency and accuracy?
- **RQ4:** Does hybrid (lexical + dense) retrieval combined with dialect mapping outperform either method alone on queries containing domain-critical rare entities?

The main contributions of this paper are:

- Identifying a specific, underaddressed gap in speech-native agricultural retrieval—dialect variation combined with domain-critical rare vocabulary—not addressed by existing speech-native retrieval systems [6], [7].
- A dialect-to-scientific-entity mapping layer bootstrapped from low-cost, publicly available resources rather than new dialect speech collection.
- A confidence-gated RAG architecture that escalates low-confidence queries to a human expert.
- A system design and experimental protocol scoped to consumer-grade, GPU-constrained hardware, reserving cloud GPU resources only for one-time adapter training.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the proposed architecture; Section IV describes the experimental protocol designed to evaluate it; Section V outlines the evaluation plan and expected outcomes for each research question, since that protocol has not yet been executed on the target dialect; Section VI reports real, measured preliminary results from a runnable software prototype of the architecture on demo-scale data, distinct from Section V's still-open hypotheses; and Section VII concludes.

II. Literature Review

A. AI-Driven Agricultural Advisory Systems

A growing body of work applies retrieval-augmented generation to agricultural advisory. Farmer.Chat [1] is the most widely cited system, deployed across Kenya, India, Ethiopia, and Nigeria and engaging over 15,000 farmers across more than 300,000 queries; it supports more than 40 crops with multilingual, multimodal (text, voice note, image) input, ingesting structured and unstructured sources into a dynamic knowledge base. KrishokBondhu [2] is architecturally the closest system to the present proposal: a phone-call-based, fully voice-in/voice-out RAG pipeline for Bengali farmers in which OCR-digitized agricultural handbooks are vectorized, ASR transcribes the call, a vector database retrieves relevant passages, a small language model generates a grounded response, and text-to-speech delivers the answer; in pilot evaluation it produced high-quality answers for 72.7% of queries and outperformed the KisanQRS benchmark [14] by 44.7% on a composite quality score. SukhaRakshak AI [3] integrates national language-AI infrastructure to deliver drought advisories in more than twenty Indian languages. AgroLLM [15] describes a simpler FAISS-based RAG chatbot for general agricultural question answering. Most recently, Sawant et al. [16] evaluate a RAG framework over package-of-practices documents for five crops across four different LLMs, reporting retrieval and generation-quality metrics (precision@k, recall@k, MRR, NDCG, BLEU, ROUGE, BERTScore); like the systems above it is text-only, with no ASR, dialect handling, or confidence-gated escalation.

Three further systems are close enough in scope to merit individual discussion. A.A.H.A.R. [17] combines an offline, quantized LLM with an ASR component fine-tuned specifically for agricultural vocabulary and regional accents, reporting sub-2-second response times and high user satisfaction in its "Aahar" app; however, it remains a cascaded ASR-then-text architecture — its ASR is adapted to dialects, but retrieval still depends on the resulting transcript, and it includes neither speech-native retrieval nor confidence-gated escalation. Krishi Sathi [18] is an intent-aware, multi-turn agricultural question-answering chatbot with ASR and TTS for accessibility in English and Hindi, reporting 97.53% query response accuracy and sub-6-second response times, but follows the same ASR-then-retrieval design and does not address dialect variation specifically. A cross-lingual RAG case study for Bengali agricultural advisory [19] is presently text-only and already introduces a keyword-injection strategy to bridge colloquial farmer terminology with scientific nomenclature — a text-domain analog of this paper's proposed dialect-mapping layer — but its own discussion explicitly names ASR integration and dialect-aware normalization of spoken queries as future work it has not yet undertaken, independent confirmation from within the same problem space that this combination remains open.

None of the systems surveyed here treats ASR-induced retrieval failure on dialect-specific or domain-critical terminology as a first-class design problem to be solved by restructuring retrieval; the closest, A.A.H.A.R. and Krishi Sathi, address dialect variation only by adapting the ASR component itself.

B. Speech Processing for Low-Resource Indian Languages and Dialects

Benchmarking work on ASR for Indian-language agricultural speech [4] finds that standard word error rate (WER) under-penalizes errors on domain-critical terms, proposing an agriculture-weighted metric that shows the dominant ASR confusion pairs across languages involve crop, chemical, and pest or disease vocabulary specifically. A complementary study on cross-lingual ASR transfer for low-resource Indic dialects [5] quantifies how code-mixing and orthographic variation degrade transcription accuracy for dialect varieties such as Garhwali, building on national speech-data initiatives including Bhashini, Vakyansh, and IndicWav2Vec, and notes that no prior work had trained or evaluated an ASR model for that dialect at all. Dialect-specific speech corpora, such as a 250-hour Chhattisgarhi ASR dataset [20] produced under the RESPIN initiative and a dedicated Malayalam agricultural speech corpus, show that domain- and dialect-specific resources exist for some Indian languages but remain incomplete in coverage — most Indian dialects, including the one targeted in this paper, have no comparable resource yet.

C. Speech-Native Retrieval

SpeechRAG [6] fine-tunes an adapter that projects speech-encoder representations into the embedding space of a frozen text retriever, enabling direct retrieval of audio passages from text or speech queries without an ASR step; retrieval experiments on spoken question-answering benchmarks show this approach matches or exceeds an ASR-then-text-retrieval cascade. Google's S2R [7] performs an analogous dual-encoder mapping in a production voice-search system and is accompanied by the Simple Voice Questions benchmark spanning seventeen languages. Underlying both approaches are self-supervised speech representation models such as wav2vec 2.0 [21] and HuBERT [22], and the broader "textless NLP" literature [23], which has found that representations optimized for ASR accuracy are not necessarily optimal for downstream spoken-language tasks. Multilingual speech-translation foundation models such as SeamlessM4T [24] provide an additional source of pretrained multilingual speech encoders relevant to adapter-based approaches. Critically, neither SpeechRAG nor S2R has been evaluated under dialect variation or domain-critical rare-entity retrieval, and neither targets agriculture as an application domain — both are general-purpose systems whose published evaluations use standard, well-resourced language varieties.

D. Retrieval Strategy: Keyword, Dense, and Hybrid Search

Comparative studies of retrieval strategy [25] consistently find that dense (semantic) retrieval outperforms lexical retrieval on paraphrased queries with little lexical overlap with the target document, while lexical methods such as BM25 retain an advantage on exact, rare, domain-specific entities — a category that includes pesticide and disease names directly relevant to agricultural queries. Hybrid retrieval, fusing lexical and dense signals, has been shown to outperform either method alone in domain-specific question answering. This is a practically important finding for the present proposal: a naive "speech embeddings beat keywords" framing would risk a result where keyword matching wins precisely on the safety-critical query category (pesticide and pest names), making hybrid retrieval — rather than a strict either/or comparison — the more defensible retrieval backbone.

E. Dialect-Aware and Colloquial-to-Formal Knowledge Representation

Agricultural knowledge graphs are a mature area of work, almost entirely built from standardized written text. Crop GraphRAG [8] constructs a multi-relational knowledge graph spanning 28 crop categories, with retrieval performed jointly over graph subgraphs and text passages. The AgCNER dataset [26] supports named-entity recognition for agricultural pests and diseases at scale. At the standards level, the EPPO ontology and related efforts [9] provide cross-system interoperable codes for plant and pest entities.

Closer to the present proposal's dialect-mapping layer, two recent works address the gap between informal farmer language and formal agricultural terminology — but neither in speech. Liu et al. [27] propose a soft-prompt-tuning method that extracts agricultural entities from colloquial, unstructured symptom descriptions using a pretrained agricultural language model, queries an agricultural knowledge graph for the matched entities, and classifies the pest or disease from the combined representation, reporting that conventional classifiers degrade substantially on farmers' natural, fuzzy language. The Bengali cross-lingual RAG case study [19] takes a simpler approach, injecting domain-specific keywords to bridge colloquial Bengali terms with scientific English nomenclature before retrieval. Both confirm that the colloquial-to-formal mapping problem is real, actively studied, and tractable — but both operate on written text, leaving the speech-domain version of this same mapping problem, applied to retrieval rather than classification, open.

F. Confidence Estimation in RAG

Dependable RAG work [10] categorizes hallucination detection into closed-book methods, which rely on internal model signals, and open-book methods, which compare model behavior with and without retrieved context. Confidence-gated abstention has been argued for explicitly in high-stakes RAG deployments in clinical decision support and financial advisory, where the cost of a confident wrong answer exceeds the cost of abstaining [11]. However, naive entropy-based confidence signals have been shown to misfire in both directions, flagging correct, well-grounded answers as uncertain due to interactions between attention mechanisms and entropy computation [12], indicating that a confidence-gating layer requires validation against labeled correct/incorrect examples rather than being assumed reliable by construction.

G. Offline and Edge Deployment of Language Models

Quantization and on-device deployment of large language models is an active systems research area. Reported figures for 4-bit quantized Llama-3 8B indicate usable but slow inference (2–3 tokens/second) on smartphones with at least 6 GB of RAM, with 8-bit variants requiring a discrete GPU with at least 8 GB of VRAM for comparable throughput [13]. Within agriculture specifically, A.A.H.A.R. [17] and the Bengali cross-lingual RAG case study [19] both report fully local, consumer-hardware deployments with sub-20-second end-to-end latency, demonstrating that offline agricultural RAG is already practically achievable for text-based systems; whether this holds once a speech-native retrieval and dialect-mapping layer are added is one of this paper's open questions (RQ3).

A recent offline plant-disease detection system deployed in rural Tigray [28] demonstrates the broader deployment pattern — quantization, on-device runtime conversion, and dialect-localized interfaces — though for a vision rather than a speech modality.

H. Synthesis

Taken together, the systems reviewed above each address a subset of the requirements motivated in Section I, but none addresses all of them jointly. Farmer.Chat, KrishokBondhu, SukhaRakshak AI, A.A.H.A.R., and Krishi Sathi demonstrate that voice-enabled agricultural RAG is deployable at scale, but all rely on an ASR-then-retrieval cascade; A.A.H.A.R. and Krishi Sathi only adapt the ASR component to dialects rather than removing the dependence on transcription. SpeechRAG and S2R demonstrate that speech-native retrieval is viable and can outperform the cascade, but neither has been evaluated under dialect variation or domain-critical rare vocabulary, and neither targets agriculture. The colloquial-to-formal mapping work of Liu et al. and the Bengali RAG case study show that normalizing informal terminology is both necessary and actively studied, but only in text. Confidence-gated abstention and offline deployment are each independently established techniques, including within agricultural RAG specifically. The combination of all four — speech-native, dialect-aware, confidence-gated, offline agricultural retrieval — does not appear in any system located during this review, and one of the closest text-based systems in the same problem space explicitly names this combination as unaddressed future work of its own [19].

III. Proposed Architecture

A. Design Overview

The architecture proposed here integrates the four components motivated in Section I into a single pipeline, illustrated in Fig. 1. A spoken farmer query is first mapped directly into a retrieval-ready representation without an intervening transcription step (Section III-B); the resulting representation is normalized against a dialect-to-scientific-entity mapping layer so that regional and colloquial terms resolve to the same retrieval targets as their standardized equivalents (Section III-C); retrieval itself is performed by a hybrid lexical-dense retriever, chosen specifically because neither method alone is reliable across both paraphrased and rare-entity queries (Section III-D); the retrieved evidence and a confidence score are then passed through a gate that either proceeds to answer generation or escalates the query to a human expert (Section III-E); and every stage except one-time adapter training is required to execute fully offline on consumer-grade hardware (Section III-F). Each subsection below states the design decision, the prior work it builds on, and the specific gap in that prior work the decision is intended to close.

B. Speech-Native Retrieval Backbone

Following SpeechRAG [6] and Google's S2R [7], the pipeline fine-tunes a lightweight adapter that projects representations from a pretrained speech encoder — wav2vec

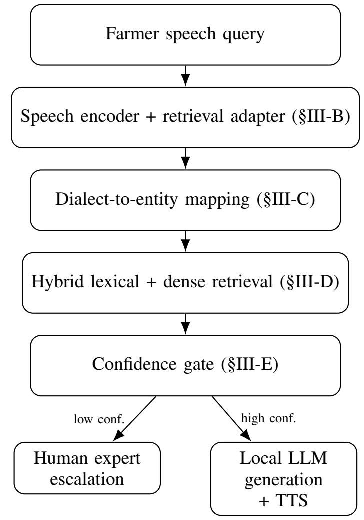

Fig. 1. Proposed pipeline. All stages except one-time adapter training (§III-F) run fully offline on consumer-grade hardware.

2.0 [21], HuBERT [22], or a multilingual encoder such as SeamlessM4T [24] where the target dialect lacks dedicated pretraining data — into the embedding space of a frozen text retriever, extending the dense-retrieval formulation of Karpukhin et al. [29] to operate on speech rather than text queries. This adapter is the only component in the pipeline permitted to use cloud GPU resources, and only once, at training time (contribution 4); at inference time it runs locally alongside the rest of the pipeline. Unlike the evaluations reported for SpeechRAG and S2R, the adapter here is trained and evaluated with explicit attention to two conditions neither prior system tested: queries containing domain-critical rare entities (pesticide, pest, and disease names) and queries spoken in dialectal or code-mixed variants of the target language, motivated by evidence that both conditions independently degrade ASR transcription accuracy [4], [5] and are expected to compound when they co-occur, consistent with the "textless NLP" premise that representations optimized for transcription accuracy are not necessarily optimal for downstream retrieval [23].

C. Dialect-to-Scientific-Entity Mapping Layer

Retrieval collapses if a normalized speech embedding still fails to align with a document indexed under standardized nomenclature. The mapping layer addresses this by bootstrapping a dialect-term-to-scientific-entity lexicon from existing public resources — agricultural knowledge graphs such as Crop GraphRAG [8], named-entity resources such as AgCNER [26], and standards-level ontologies such as EPPO [9] — rather than through new dialect speech collection, consistent with contribution 2. This follows the same motivation as the soft-prompt-tuning approach of Liu et al. [27], which showed that conventional classifiers degrade substantially on farmers' colloquial symptom descriptions, and the keyword-injection strategy used in the Bengali cross-lingual RAG case study [19]; both operate on written text, whereas the mapping layer proposed here is designed to operate on the speech-embedding representation produced in Section III-B, since that system's own future-work discussion identifies dialect-aware normalization of spoken queries as unresolved. Two integration strategies are considered: a post-ASR text-normalization path, retained only as a fallback because it reintroduces a transcription dependency for this one layer, and an embedding-space mapping path that aligns dialect-term speech embeddings directly against the retriever's embedding space, preferred because it requires no transcription step anywhere in the pipeline.

D. Hybrid Retrieval Strategy

Comparative retrieval studies show dense retrieval outperforming lexical retrieval on paraphrased queries with low lexical overlap, while lexical methods such as BM25 retain the advantage on exact, rare, domain-specific entities [25] — precisely the pesticide and disease name category this paper treats as safety-critical. The architecture therefore fuses lexical and dense signals rather than committing to either alone, so that a colloquial, paraphrased query and an exact rare-entity query are both served by the retrieval mode best suited to it. This choice is deliberately positioned against a naive "speech embeddings beat keywords" framing: since a purely dense speech-native retriever risks losing precisely the safety-critical query category to a simpler keyword baseline, hybrid retrieval is treated as the default backbone to be evaluated, not a strict either/or alternative to speech-native retrieval.

E. Confidence-Gated Escalation

Because an incorrect agricultural recommendation can damage a season's crop, the architecture inserts a confidence gate between retrieval and answer generation, following the case made for confidence-gated abstention in comparably high-stakes clinical and financial RAG deployments [11]. The gate combines closed-book signals (internal model confidence) and open-book signals (divergence between model behavior with and without retrieved context), following the taxonomy of Dependable RAG [10], rather than relying on a single score. This design choice is directly informed by a documented failure mode: naive entropy-based confidence signals have been shown to misfire in both directions, flagging correct, well-grounded answers as uncertain due to interactions between attention mechanisms and entropy computation [12]. Accordingly, the gate is treated as a component requiring empirical validation against labeled correct/incorrect examples (Section IV-E) rather than a mechanism assumed reliable by construction. Queries falling below the confidence threshold are escalated to a human expert rather than answered, mirroring the human-in-the-loop fallback already present in call-center-style deployments such as KrishokBondhu [2].

F. Offline Deployment Scoping

All pipeline stages other than the one-time speech-retrieval adapter training (Section III-B) are required to run without a live network connection, on consumer-grade, GPU-constrained hardware. This mirrors the deployment pattern already demonstrated for text-only offline agricultural RAG systems such as A.A.H.A.R. [17] and the Bengali cross-lingual RAG case study [19], and for a vision-modality offline system in rural Tigray [28], and is bounded by reported feasibility figures for quantized on-device language models — 2–3 tokens/second for 4-bit Llama-3 8B on a smartphone with at least 6 GB RAM, with 8-bit variants requiring a discrete GPU with at least 8 GB VRAM for comparable throughput [13]. Whether these figures still hold once a speech encoder, adapter, dialect-mapping layer, and confidence gate are added on top of the language model is treated as an open question (RQ3) evaluated in Section IV, not an assumption carried over from text-only systems.

IV. Experimental Protocol

A. Overview

No experiments have yet been executed; this section specifies the protocol by which the architecture in Section III would be evaluated against RQ1–RQ4, so that the evaluation plan in Section V can be read as a set of falsifiable hypotheses rather than an assumed outcome.

B. Query Dataset Construction

The evaluation requires a query set stratified along two axes motivated in Sections I and III: standard-terminology versus dialectal/colloquial phrasing, and common versus domain-critical rare-entity queries (pesticide, pest, and disease names). Prior characterizations of real farmer helpline query volume and content [30], [31] inform the query distribution the dataset should approximate, so that the stratified test set is representative of actual query patterns rather than an artificially uniform sample. Where dedicated dialect speech resources exist, such as the RESPIN Chhattisgarhi corpus [20], they are used directly; where the target dialect has no comparable resource, a small evaluation set is collected specifically for the rare-entity and dialectal strata, consistent with the constraint that no large new dialect speech corpus is collected for training (Section III-C).

C. Baselines and Conditions

Four retrieval conditions are compared under identical query sets, isolating the specific comparisons RQ1 and RQ4 require:

1) **ASR-cascaded retrieval** — a Whisper-based [32] transcription step followed by text retrieval, representing the dominant existing design pattern (Farmer.Chat, KrishokBondhu, A.A.H.A.R., Krishi Sathi).
2) **Keyword-only (lexical) retrieval** — BM25 over ASR transcripts, isolating lexical retrieval performance in the presence of transcription error.
3) **Dense semantic retrieval** — text-embedding retrieval following the dense passage retrieval formulation [29], applied both to ASR transcripts and, where applicable, to speech-native embeddings from Section III-B.
4) **Hybrid retrieval** — the fused lexical-dense condition of Section III-D, evaluated both with and without the

dialect-to-entity mapping layer of Section III-C, isolating the mapping layer's contribution (RQ2).

All conditions retrieve from the same underlying knowledge base, constructed following the retrieval-augmented generation formulation of Lewis et al. [33], so that differences in measured accuracy are attributable to the retrieval mechanism rather than to differences in the underlying corpus.

D. Evaluation Metrics

Retrieval quality is measured with standard recall@k and mean reciprocal rank, reported separately for the standard-terminology and dialectal/rare-entity strata rather than as a single aggregate figure, following the rationale that aggregate metrics can mask exactly the failure mode this paper targets — the same rationale underlying the agriculture-weighted WER metric proposed for ASR evaluation [4]. End-to-end latency is measured from spoken query to delivered answer on the target consumer-hardware configuration (Section IV-F). Confidence-gate quality is measured separately from retrieval quality, via calibration error and via abstention precision/recall against a labeled set of known-correct and known-incorrect answers, directly addressing the INTRYGUE-documented risk that confidence signals can misfire in both directions [12].

E. Confidence Gate Validation Protocol

Because a miscalibrated gate can be worse than no gate at all [12], the gate is validated on a held-out labeled set before being used to report any end-to-end system result: candidate confidence signals (closed-book and open-book, per [10]) are scored against ground-truth correctness labels, and a signal or combination of signals is adopted for the full pipeline only if it exceeds a pre-registered calibration threshold on this held-out set.

F. Hardware Target

All inference-time components (Sections III-B through III-E) are benchmarked on a single, explicitly stated consumer-hardware configuration — a GPU-constrained laptop or mini-PC class device, consistent with the RAM/VRAM figures in [13] — rather than left as an unspecified "consumer-grade" claim, so that the latency and accuracy figures reported in Section V are reproducible against a fixed target.

V. Evaluation Plan and Expected Outcomes

As the protocol in Section IV has not yet been executed, this section states, for each research question, the comparison that would answer it and the pattern of results that would confirm or disconfirm the corresponding hypothesis — not observed data.

A. RQ1 — Speech-Native Retrieval Under Dialect and Rare-Entity Conditions

Hypothesis: speech-native retrieval (Section III-B) matches or exceeds ASR-cascaded retrieval in aggregate, consistent with prior results for SpeechRAG and S2R [6], [7], but the size of that advantage is expected to *grow* on the dialectal and rare-entity strata specifically, since those are exactly the conditions under which ASR transcription error concentrates [4], [5]. **Disconfirming pattern:** if the speech-native advantage shrinks or reverses on the dialectal/rare-entity strata relative to the standard-terminology stratum, this would indicate the adapter itself fails to generalize to dialect variation, motivating dialect-specific adapter fine-tuning as future work rather than confirming RQ1 as stated.

B. RQ2 — Dialect-to-Entity Mapping Contribution

Hypothesis: hybrid retrieval with the mapping layer (Section III-C) outperforms hybrid retrieval without it specifically on the dialectal/colloquial query stratum, with little or no difference on the standard-terminology stratum, isolating the mapping layer's contribution to exactly the queries it targets. **Disconfirming pattern:** a uniform improvement across both strata would suggest the mapping layer is acting as a general retrieval-quality improvement rather than a dialect-specific one, which would still be a positive result but would not confirm RQ2 as a dialect-targeted claim.

C. RQ3 — Offline Feasibility on Consumer Hardware

Hypothesis: the full pipeline meets a pre-specified latency budget (informed by the sub-2-second to sub-20-second range reported for text-only offline agricultural systems [17], [19], adjusted upward for the added speech and mapping components) on the hardware target specified in Section IV-F, without a significant accuracy regression relative to a cloud-hosted configuration of the same pipeline. **Disconfirming pattern:** if latency exceeds the budget or accuracy regresses substantially under quantization, this would indicate the current architecture requires either further compression or a revised hardware target, directly answering RQ3 in the negative rather than leaving it unresolved.

D. RQ4 — Hybrid Retrieval with Dialect Mapping vs. Either Alone

Hypothesis: hybrid retrieval combined with dialect mapping outperforms lexical-only and dense-only retrieval individually on the rare-entity stratum, where prior comparative work predicts lexical and dense methods trade off [25], confirming that neither method alone is sufficient once dialect variation and rare entities co-occur. **Disconfirming pattern:** if one retrieval mode alone matches hybrid performance even on the rare-entity stratum, this would indicate the hybrid fusion adds complexity without a corresponding accuracy benefit for this query population.

E. Threats to Validity

The evaluation plan above depends on dialect-stratified query data whose scale is inherently limited by the absence of large existing dialect speech resources for the target dialect (Section IV-B), which constrains statistical power on exactly the strata RQ1 and RQ2 depend on; this limitation, and mitigations such as bootstrapped or synthetic dialectal query augmentation, are discussed further in Section VII.

VI. Prototype Implementation and Preliminary Measurements

A. Scope and Purpose of This Section

Sections III–V describe an architecture and a protocol for evaluating it, and state explicitly that the protocol had not been executed. Since that text was written, a best-effort, genuinely runnable software prototype of the architecture in Section III has been built and exercised end to end, and this section reports the real, measured numbers that prototype produces. This is not the same thing as executing the Section IV protocol: the prototype runs on small, hand-curated, demo-scale data rather than the dialect-stratified corpus Section IV-B calls for, and — because no recorded speech corpus for the target dialect exists, which remains this paper's central open problem — every audio-involving measurement below uses `espeak-ng` text-to-speech synthesis of query text rather than native dialect speech. The results in this section should therefore be read as an engineering stress-test of the mechanism proposed in Section III: evidence that each component runs, composes, and behaves sensibly, not as an answer to RQ1–RQ4 as posed. Two follow-up measurements (Sections VI-F and VI-H) were added in a later revision to specifically close two items this section had originally left as future work — the confidence gate's open-book signal weakness and the untested effect of LLM quantization — and are reported alongside, not in place of, the original findings. Section V's hypotheses remain open; Section VI-I states precisely what this prototype does and does not change about that.

B. Prototype System and Demo Data

The prototype implements all four pipeline stages of Fig. 1: a hybrid BM25 + dense retriever (Reciprocal Rank Fusion, $k = 60$) over a 42-passage hand-authored agricultural knowledge base spanning common and domain-critical-rare pest, disease, and pesticide entries; a 29-entry dialect-to-entity lexicon bootstrapped, per the contribution-2 constraint, from generally known North/Central-Indian farmer-extension terminology rather than a domain-expert-reviewed or public-KG-sourced resource — a demo-scale stand-in for the resource contribution 2 specifies, not that resource itself; a speech-native retrieval adapter (frozen wav2vec2 [21] encoder plus a small trained projection head, following [6], [7]) trained with an in-batch InfoNCE contrastive objective; an ASR-cascaded baseline using `faster-whisper` "tiny" [32]; and a confidence gate combining a closed-book retrieval-margin signal with an open-book lexical-grounding-overlap signal, per the taxonomy of [10]. Generation uses Qwen2.5-0.5B-Instruct in fp16, falling back to a deterministic template when the LLM is unavailable. All measurements below were run on a single NVIDIA RTX 3050 Laptop GPU (6 GB VRAM), stating a concrete hardware target as Section IV-F requires. Evaluation uses 50 queries stratified 2×2 over standard/dialectal phrasing and common/domain-critical-rare entities, matching Section IV-B's stratification design.

C. Retrieval Comparison

Table I reports recall@1, by query stratum, for six retrieval conditions. Conditions 1–4 take the ground-truth query *text* directly, isolating the retrieval-strategy and dialect-mapping backbone from any ASR- or speech-encoding-induced error; only conditions 5 and 6 take synthesized speech *audio* as input, which is the only fair test of this paper's speech-native premise, and — as discussed below — neither currently performs well. This distinction matters for how the table should be read: the strong condition-4 numbers below demonstrate the retrieval-and-mapping backbone works given correct input text, not that the pipeline retrieves well from speech; Sections VI-D and VII return to why no speech-input condition yet reaches comparable performance. Dialect mapping is the single largest lever measured among the text-input conditions: every condition without it scores at or near zero recall@1 on both dialectal strata, while adding it (condition 4) raises dialectal recall@1 to 0.615 — a real, clean, demo-scale confirmation of the RQ2/RQ4 premise that a mapping layer is specifically what unlocks dialectal queries, though the 42-passage KB used here has fairly distinct inter-passage vocabulary, and the same experiment on a larger, messier knowledge base would need to be re-run before trusting this exact figure. The ASR-cascaded baseline (condition 5) is badly degraded by dialectal queries (0.000 recall@1 on both dialectal strata) and, notably, underperforms text-only hybrid retrieval even on standard-phrasing queries (0.583 vs. 0.917 on standard/common), because Whisper "tiny" mistranscribes even the synthetic English-voice audio substantially; this is consistent with the ASR-error concentration reported by [4], [5], but should be read as a property of this specific small model and synthetic-audio pipeline, not a general claim about Whisper. (A same-day re-run of this evaluation reproduced every other cell in this table exactly and this condition's dialectal/common cell specifically at 0.000 rather than an earlier 0.077 — one query's transcription flipping between correct and incorrect, a known run-to-run non-determinism of `faster-whisper` on GPU stemming from non-deterministic cuDNN kernel selection; the qualitative finding is unchanged.) The trained speech-native adapter (condition 6) scores 0.020 recall@1 overall on this held-out eval set — effectively chance for a 42-way retrieval problem — despite demonstrably learning a real retrieval signal on its own training distribution (Section VI-D) — a genuine, and informative, negative result rather than an implementation defect.

D. Speech-Native Adapter: Mechanism Works, Generalization Does Not

Trained for 100 epochs (fixed seed, $\sim$20 s wall time) with an in-batch InfoNCE contrastive loss on 108 auto-generated English template sentences, the adapter reaches 61.1% top-1 passage retrieval accuracy on its own 18-example held-out validation split — roughly 26× the 2.4% chance level for this 42-way retrieval problem, and a real, positive demonstration that the frozen-speech-encoder-plus-small-adapter mechanism of [6], [7] works end to end without any transcription step.

TABLE I

RECALL@1 BY RETRIEVAL CONDITION AND QUERY STRATUM (PROTOTYPE, n=50 DEMO EVAL QUERIES; SEE SECTION VI-A FOR SCOPE CAVEATS). CONDITIONS 1–4 TAKE GROUND-TRUTH QUERY TEXT AS INPUT; ONLY CONDITIONS 5–6 TAKE SYNTHESIZED SPEECH AUDIO, THE ONLY FAIR TEST OF THE PAPER'S SPEECH-NATIVE PREMISE.

Condition	All	Std./Common	Std./Rare	Dial./Common	Dial./Rare
1. Keyword only (BM25) [text input]	0.500	0.917	1.000	0.077	0.077
2. Dense only [text input]	0.400	0.833	0.833	0.000	0.000
3. Hybrid, no dialect mapping [text input]	0.460	0.917	1.000	0.000	0.000
4. Hybrid + dialect mapping [text input]	**0.780**	0.917	1.000	**0.615**	**0.615**
5. ASR cascade + hybrid + dialect mapping [audio input]	0.380	0.583	1.000	0.000	0.000
6. Speech-native adapter, no dialect mapping [audio input]	0.020	0.000	0.000	0.000	0.077

On the held-out evaluation query set's full natural-language symptom descriptions and romanized dialectal queries — a materially different phrasing distribution from the short templates it trained on — recall@1 falls to 0.020 overall (Table I, condition 6), effectively at chance. This generalization gap is exactly the kind of result RQ1 anticipates as an open question rather than assumes resolved: the mechanism transfers, a small synthetic training signal does not, and dialect-speech generalization specifically remains untested by this prototype, since no dialect speech data exists to test it on.

E. Confidence Gate Validation

Following the Section IV-E protocol, the gate was validated on 50 labeled examples (39 correct / 11 incorrect; unconditional accuracy 0.780) before being credited with any end-to-end result. Table II shows the threshold sweep: at threshold 0.5, the gate answers 32% of queries at 0.875 accuracy (vs. the 0.780 always-answer baseline) while escalating 81.8% of the actually-wrong answers in the set. This clears the minimal bar a confidence gate must clear. Validation also surfaced a real limitation, exactly the kind [12] warns naive confidence signals can hide: the open-book grounding-overlap signal varies only from 0.868 to 0.937 across every example in this run, correct or not, because the templated answers used to build this labeled set quote their own top-1 passage verbatim — so the answer is "grounded" by construction even when that passage is wrong. Almost all of the gate's discriminative power in this run therefore comes from the closed-book retrieval-margin signal alone, not the closed-book/open-book combination the design in Section III-E intended (Expected Calibration Error over 5 bins: 0.291). This is the validation step named in Section IV-E doing precisely what it is for: catching, before deployment, a signal that looks reasonable in aggregate but is silently carried by only one of its inputs. A production version of the gate would need this signal re-validated against non-template-quoting LLM answers, where grounding overlap can actually diverge from correctness.

F. Confidence Gate Re-Validation with Real LLM Answers

Section VI-E's validation used templated answers that quote their retrieved passage verbatim — itself flagged there as an artificially easy case for the open-book grounding-overlap signal, since such an answer is grounded by construction regardless of whether the retrieved passage was even correct.

TABLE II

CONFIDENCE GATE THRESHOLD SWEEP (PROTOTYPE, n=50 LABELED EXAMPLES)

Thresh.	Coverage	Acc. if ans.	Esc. prec.	Esc. recall
0.4	1.000	0.780	n/a	0.000
0.5	0.320	0.875	0.265	0.818
0.6	0.000	n/a	0.220	1.000

To test whether the resulting low-variance finding was a genuine property of the signal or an artifact of that templated construction, the same 50 eval queries were re-labeled using real Qwen2.5-0.5B-Instruct generation (the same grounded-generation prompt used elsewhere in the pipeline, Section VI-B) in place of the template, with correctness still defined identically — whether the retrieved top-1 passage matched the gold id, a property of retrieval rather than of the answer text, so the two label sets remain directly comparable. All 50 queries produced a real LLM answer (zero template fallbacks), and the result was identical across two independent re-runs (deterministic greedy decoding).

Table III compares the two label sets. Against real answers, `grounding_overlap` ranges from 0.000 to 1.000 (mean 0.640) rather than the templated set's near-constant 0.868–0.937 — confirming the original low-variance finding was specifically an artifact of the template-quoting construction, not an inherent property of the signal. The gate's discriminative performance also improves on this set: at its best threshold (0.4, chosen by the same $n_{\text{answered}} \geq 5$ selection rule as Section VI-E), it answers 34% of queries at 0.941 accuracy against the same 0.780 always-answer baseline — a +0.161 margin, larger than the templated set's +0.095 — while catching 90.9% of actually-wrong answers (vs. 81.8%). Calibration (Expected Calibration Error) is nominally worse on this set (0.425 vs. 0.291), but this follows mechanically from the signal now taking a much wider range of values rather than indicating worse gate behavior in the accuracy/coverage sense that matters for the deployment decision. This closes one of the two open problems Sections VI-E and VII originally named as future work: on this evidence, the open-book signal's apparent weakness was a limitation of the first validation set's construction, not (so far) an inherent flaw in the signal design — though, per [12], this remains a 50-example demo-

TABLE III
CONFIDENCE GATE: TEMPLATED-ANSWER LABELS (BEST THRESH. 0.5) VS. REAL-LLM-ANSWER LABELS (BEST THRESH. 0.4); PROTOTYPE, n=50

Metric	Templated	Real LLM
Coverage	0.320	0.340
Acc. if answered	0.875	0.941
Margin vs. baseline	+0.095	+0.161
Escalation recall	0.818	0.909
Escalation precision	0.265	0.303
Overlap range	0.868–0.937	0.000–1.000
ECE (5 bins)	0.291	0.425

TABLE IV
STEADY-STATE PER-STAGE LATENCY, TEXT-QUERY PATH (PROTOTYPE, RTX 3050 LAPTOP 6GB, n=20)

Stage	Mean	p95
Dialect mapping	0.1 ms	0.6 ms
Hybrid search (BM25+dense+fusion)	3.8 ms	4.3 ms
Confidence gate scoring	0.0 ms	0.1 ms
LLM generation ($\leq$120 tok)	1282.5 ms	1476.4 ms

scale check on synthetic-TTS-derived text, not a deployment certification, and should be re-validated again at larger scale before being trusted operationally.

G. Offline Latency and Memory (RQ3)

Table IV reports steady-state per-stage latency on the RTX 3050 Laptop target. Dialect mapping, hybrid retrieval, and the confidence gate are each sub-10 ms and effectively free next to generation; LLM generation dominates end-to-end latency at $\sim$1.3 s mean for a 120-token-capped response. Peak VRAM with every model loaded simultaneously — dense retriever, Whisper tiny, wav2vec2 plus adapter, and the fp16 LLM — measured 1.54 GB, comfortably inside the 6 GB budget with $\sim$4.5 GB headroom. This lands in roughly the same sub-2-second ballpark reported for a comparable offline agricultural LLM system on comparable hardware [17], which is consistent with, though not proof of, RQ3's premise holding once the speech, dialect-mapping, and confidence-gating layers are stacked on top of generation — for this demo-scale, 42-passage KB. The LLM here runs in fp16; Section VI-H reports what happens when INT4/INT8 quantization is actually applied to this exact pipeline, which turns out to qualify rather than confirm the throughput expectations set by [13]. Retrieval latency over a production-scale KB (thousands of passages) also should not be extrapolated linearly from this 42-passage result.

H. Quantization Benchmark: An Unexpected Latency Regression

Section VI-G's latency figures use the LLM in fp16 and explicitly leave quantization — the mechanism [13] reports as enabling practical on-device LLM inference at all — untested against this exact pipeline. To close that gap, the generation

TABLE V
LLM GENERATION STAGE ALONE: FP16 VS. INT8 VS. INT4 (PROTOTYPE, RTX 3050 LAPTOP 6GB, n=20 PER MODE, ISOLATED SUBPROCESS PER MODE)

Mode	Mean	p95	Peak VRAM
fp16 (baseline)	1176.3 ms	1343.7 ms	0.95 GB
INT8	6752.9 ms	8592.3 ms	0.61 GB
INT4	1755.0 ms	2401.5 ms	0.47 GB

stage alone (Qwen2.5-0.5B-Instruct) was re-benchmarked under INT8 and INT4 loading via `bitsandbytes`, each mode measured in its own fresh process on the same RTX 3050 Laptop GPU so that peak VRAM and latency are attributable to that mode alone rather than compounded across modes (Table V).

Quantization reduces peak VRAM as expected for the LLM alone (fp16 0.95 GB $\rightarrow$ INT8 0.61 GB $\rightarrow$ INT4 0.47 GB), but — contrary to the throughput improvements [13] reports for quantized inference on smartphone-class hardware — both quantized modes are substantially *slower* than fp16 on this GPU. INT8 averaged 6.1–6.8 s per query across three independent runs (roughly 5–6$\times$ slower than fp16's consistent $\sim$1.2 s), and INT4 averaged 1.6–1.8 s (40–50% slower). This is a genuine, informative negative result, not an implementation defect: `bitsandbytes`' INT8 path relies on an outlier-decomposition matmul whose overhead is known to dominate at small batch sizes and short sequence lengths on consumer (non-datacenter) GPU kernels, and a 0.5B-parameter model generating at most 120 tokens is close to a worst case for that overhead to be amortized away. This directly qualifies RQ3: the offline-deployment feasibility Section VI-G reports holds specifically for the fp16 configuration actually benchmarked there — naively enabling INT8/INT4 quantization on this pipeline, on this GPU class, would make end-to-end latency *worse*, not better, reversing the intuition that quantization is an unconditionally beneficial lever for offline deployment. Whether this reverses at larger model scale (where quantized kernels have more compute to amortize their overhead against) or on datacenter-class GPUs with dedicated low-precision tensor-core paths remains untested here and is a narrower, more concrete open question than the general quantization question this section previously left unaddressed.

I. What This Prototype Does and Does Not Establish

Taken together, this prototype demonstrates that the architecture of Section III is buildable, that its four components compose into a working end-to-end pipeline (24 passing tests across retrieval, dialect mapping, the confidence gate, and full-pipeline smoke tests), and that on demo-scale data the mechanism behaves in the direction the paper's hypotheses predict for dialect mapping (RQ2) and offline latency (RQ3). It surfaces one genuine, still-open problem that a purely architectural proposal could not have discovered — speech-native retrieval's generalization gap (RQ1, Section VI-D) — and one that follow-up validation resolved rather than confirmed as a fundamental flaw: the confidence gate's apparent open-book signal weakness (Section VI-E) turned out, on re-validation against real LLM answers (Section VI-F), to be an artifact of the first validation set's template-quoting construction, with the signal showing genuine discriminative variance once that artifact was removed. A second follow-up measurement introduced a new, narrower open problem of its own: quantizing the local LLM (Section VI-H) reduces memory as expected but substantially *increases* latency on this consumer GPU, directly qualifying RQ3's offline-feasibility finding to the fp16 configuration actually benchmarked. It does not establish that any of these results hold on real dialect speech, at production knowledge-base scale, on a different GPU class, or at a larger model scale, and it does not substitute for the dialect-stratified evaluation Section IV-B specifies on the target dialect, since the underlying speech data for that evaluation still does not exist. Section V's hypotheses are therefore still open exactly as stated; this section adds a real, if small and demo-scale, existence proof that the mechanism designed to test them runs, and shows that even the same prototype's own earlier findings can be revised — in either direction — by further honest measurement rather than assumed settled once reported.

VII. Conclusion

This paper has argued that no existing agricultural advisory system combines speech-native retrieval, dialect-to-scientific-entity mapping, confidence-gated escalation, and fully offline operation, despite each being independently established: speech-native retrieval has been shown to match or exceed ASR-cascaded retrieval in general-purpose settings [6], [7]; colloquial-to-formal entity mapping has been shown to be necessary and tractable in text [19], [27]; confidence-gated abstention has been argued for in other high-stakes RAG domains [11]; and offline deployment has been demonstrated for text-based agricultural RAG specifically [17], [19]. Section III integrates these four components into a single pipeline, Section IV and Section V specify how each of RQ1–RQ4 would be tested and what pattern of results would confirm or disconfirm each hypothesis, and Section VI reports real measurements from a runnable prototype of that pipeline on demo-scale data.

The principal limitation of the present paper is that its central research questions remain empirically open on the target dialect: the prototype reported in Section VI demonstrates that the architecture of Section III is buildable and behaves, on demo-scale data, in the direction the paper's hypotheses predict for dialect mapping and offline latency in its fp16 configuration — while also surfacing a genuine unresolved problem, the speech-native adapter's failure to generalize past its training distribution (Section VI-D) — but no component has yet been evaluated on real target-dialect speech, because that corpus does not exist, and the hypotheses in Section V are therefore still predictions to be tested rather than reported findings. Committing to a falsifiable protocol (Section IV) before running experiments, and then reporting the prototype's negative and partial results in Section VI alongside its positive ones, is intended to guard against the kind of confirmation-driven evaluation that a system built and evaluated by the same process would otherwise risk. This guard was exercised twice in this paper's revision history, in opposite directions: Section VI-E's first confidence-gate validation surfaced a signal that looked reasonable in aggregate but was silently carried by only one of its two inputs, exactly the failure mode [12] warns naive confidence signals can hide, and Section VI-F's follow-up then traced that specific weakness to an artifact of the first validation set's construction rather than a property of the signal itself, improving the gate's measured performance once corrected. Conversely, Section VI-H's quantization benchmark, added expecting a latency improvement per [13], instead surfaced a latency regression that qualifies Section VI-G's offline-feasibility finding to the fp16 configuration actually measured. Reporting a finding that a later measurement partially overturns, rather than quietly revising history, is the same honesty discipline Section VI-A commits to at the outset, applied here to this paper's own later revisions as much as to its original claims.

Future work is therefore the direct execution of Section IV on real data: collection or acquisition of the dialect-stratified evaluation set described in Section IV-B for the target dialect, re-training and re-evaluating the speech-native adapter and dialect-mapping layer on that data rather than the demo corpus and synthetic TTS audio used in Section VI, and implementation of the embedding-space dialect-mapping path so that dialect mapping composes with speech-native retrieval rather than only the post-ASR text path evaluated here. Two items from an earlier version of this future-work list — re-validating the confidence gate's open-book signal against non-template-quoting LLM answers, and measuring the local LLM under quantization — have since been executed (Sections VI-F and VI-H) and are reported as findings above rather than remaining as future work; each produced a narrower follow-on question in its place: the gate's open-book signal should be re-validated again at larger scale and on real (non-synthetic-TTS-derived) queries before being trusted operationally, and the quantization latency regression found here should be re-tested at larger model scale and on datacenter-class GPUs with dedicated low-precision tensor-core paths, where the outlier-decomposition overhead identified in Section VI-H may amortize differently. Beyond executing this protocol, two extensions follow naturally from the synthesis discussion in Section II: first, extending the dialect-to-entity mapping layer beyond the single target dialect evaluated here, toward the many Indian dialects that, like the one targeted in this paper, presently have no dedicated speech or mapping resource [5], [20]; and second, examining whether the confidence-gating and offline-deployment components generalize to other high-stakes, low-connectivity domains — clinical and financial advisory — where confidence-gated RAG has been argued for but not, to this paper's knowledge, combined with speech-native retrieval either [10], [11].

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